

HA DONG

Curriculum Vitae

Cambridge, Massachusetts | hadong@fas.harvard.edu | [ORCID](#) | [Google Scholar](#) | [LinkedIn](#)

RESEARCH INTERESTS

Circuit- and algorithm-level accounts of how nervous systems produce fast, robust behavior, and how those principles inform the design of learning systems.

EDUCATION

Harvard University

Ph.D. in Neuroscience, Kempner Graduate Fellow

Cambridge, MA

Aug. 2026 – Expected May 2032

Amherst College

B.A. in Mathematics, Physics, and Neuroscience, *summa cum laude*

Amherst, MA

Aug. 2022 – May 2026

Honors Thesis: “Exploratory Analysis of Escape Behavior and Circuitry in Larval Zebrafish”

Advisor: Josef G. Trapani, Ph.D.

Harvard College

Visiting Undergraduate Students Program

Cambridge, MA

Jan. – May 2025

Coursework: Modern Neuroanatomy, Biological and Artificial Intelligence, Electron Microscopy, Graph Theory

RESEARCH EXPERIENCE

Amherst College, Department of Neuroscience

Gregory S. Call Research Fellow (*Advisor: Dr. Josef G. Trapani*)

Amherst, MA

Sep. 2025 – May 2026

- Developed a circuit model of the temporal dynamics of C-bends during larval zebrafish startle responses, integrating field potential, calcium imaging, and behavioral data for an honors thesis.
- Supported electrophysiology and imaging experiments investigating the effects of low-dose cisplatin on hair-cell function and mitochondrial morphology.

Harvard University, Center for Brain Science

Research Intern (*Advisors: Dr. Venkatesh Murthy, Dr. Jacob Zavatone-Veth*)

Cambridge, MA

Feb. – Aug. 2025

- Developed an inverse inference algorithm to extract latent finite-state controller policies and memories from behavioral data.
- Designed a behavioral paradigm and collected data to investigate how conflicting auditory and olfactory signals influence trail-following behavior.

Vanderbilt University, Department of Chemistry

Undergraduate Researcher (*Advisors: Dr. Jens Meiler, Dr. Yunchao Liu*)

Nashville, TN

May 2023 – May 2025

- Developed WelQrate, an open-source benchmark comprising nine high-throughput screening datasets, a standardized chemical-curation pipeline, and benchmarking frameworks for probabilistic, sequence-based, and graph-based machine-learning models.
- Developed and evaluated a graph-learning augmentation method integrating expert-crafted molecular descriptors into graph representations, improving predictive performance across multiple graph neural-network architectures.

Alkermes plc

R&D Intern (*Advisors: Dr. Joerg Bentzien, Dr. Polina Vanyukov, Dr. Shinichiro Wachi*)

Waltham, MA

Sep. – Dec. 2024

- Developed an end-to-end machine-learning pipeline using ChEMBL data and random-forest models to predict multiple bioactivity endpoints for serotonin (5-HT) and dopamine receptors.

California Institute of Technology

Summer Undergraduate Research Fellow (*Advisor: Dr. Markus Meister*)

Pasadena, CA

Jun. – Sep. 2024

- Helped design and engineer a closed-loop behavioral system in which artificial prey evaded hunting mice using naturalistic escape strategies.
- Collected Neuropixels and keypoint-tracking data and applied hidden Markov models to characterize internal states underlying hunting behavior and their relationship to superior colliculus activity.

PUBLICATIONS

[†]Equal contribution. [‡]Co-senior author.

Journal Articles

Liu, Y., Moretti, R., Wang, Y., **Dong, H.**, Yan, B., Bodenheimer, B., Derr, T.[‡], & Meiler, J.[‡] (2025). Advancements in ligand-based virtual screening through the synergistic integration of graph neural networks and expert-crafted descriptors. *Journal of Chemical Information and Modeling*, 65(10), 4898–4905. [DOI] [PDF]

Refereed Conference Proceedings

Liu, Y.[†], **Dong, H.**[†], Wang, X., Moretti, R., Wang, Y., Su, Z., Gu, J., Bodenheimer, B., Weaver, C. D., Meiler, J.[‡], & Derr, T.[‡] (2024). WelQrate: Defining the gold standard in small molecule drug discovery benchmarking. *Advances in Neural Information Processing Systems*, 37, 53222–53236. [DOI] [PDF] [Website]

PRESENTATIONS

Posters

Pollak, D., Wang, J., **Dong, H.**, Rios, A., Li, X. H., Zhong, E., & Meister, M. (2026, August). A parametric guidance law model of hunting in mice. Poster presented at the *9th Annual Conference on Cognitive Computational Neuroscience (CCN)*, New York, NY. [Abstract]

Talks

Dong, H., Meister, M., & Pollak, D. (2024). The states of hunting. Talk presented at the *California Institute of Technology Summer Research Seminar*, Pasadena, CA.

Dong, H., Meiler, J., & Liu, Y. (2024). Advances in drug discovery. Talk presented at the *AI in the Liberal Arts Undergraduate Conference*, Amherst, MA.

Dong, H., Meiler, J., & Liu, Y. (2023). Deep graph learning benchmark for ligand-based drug discovery. Talk presented at the *Amherst College Summer Research Seminar*, Amherst, MA.

CREATIVE WORKS

Dong, H., & Engelhardt, J. (2024). Sounds of Faith: A story of Quranic reciters in the Connecticut River Valley. *Soundscapes of the Connecticut River Valley online archive*. [Documentary film]

AWARDS, HONORS & GRANTS

2026	Kempner Graduate Fellowship	Harvard University
2026	Ryan EDGE Fellowship	Harvard Medical School
2026	Obed Finch Slingerland Memorial Prize	Amherst College Board of Trustees
2025	Charles Hamilton Houston Research Grant (\$5,000)	Amherst College
2024–2025	Break Through Tech AI Fellowship	MIT Schwarzman College of Computing
2024	Scholar Travel Award	NeurIPS

2024	Gregory S. Call Travel Award (\$1,000)	Amherst College
2024	Meiklejohn Summer Research Grant (\$5,000)	Amherst College
2023	Meiklejohn Summer Research Grant (\$5,000)	Amherst College
2022	The Grosfeld Family Scholarship	Amherst College

Pre-college honors

2021	Top 20 Outstanding Young Vietnamese Faces Award (<i>national youth honor</i>)	Government of Vietnam
2021	Labor Order, Third Class (<i>national civilian order of merit</i>)	Government of Vietnam
2021	Gold Medal, International Medicine and Disease Olympiad	IMDO Foundation Board
2021	Exemplary Young Citizen Award	Vietnam Youth Federation
2020	Top 50 Global Finalist, Global Student Prize	Chegg.org & Varkey Foundation
2020	Silver Medal, Rudolf Ortway International Competition in Physics for Undergraduates	Hungarian Association of Physics Students
2020	Silver Medal, International Biology Olympiad	IBO Committee

PROFESSIONAL DEVELOPMENT

2022–2026	Education Profession Fellowship, Loeb Career Center, Amherst College
2022–2026	Meiklejohn Fellowship, Loeb Career Center, Amherst College

TEACHING

Instructor

2019–2022	Biology Teacher, High School for Gifted Students, Vietnam National University Subjects: Molecular and Cellular Biology, Plant and Animal Physiology, Genetics, Evolutionary Biology, Ecology Notable outcomes: 1 International Biology Olympiad gold medal, 14 National Olympiad medals
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Teaching Assistant

Spring 2026	Animal Physiology (Amherst College) Instructor: Dr. Josef Trapani
Fall 2023	Electromagnetism and Optics (Amherst College) Instructors: Dr. Juan Vasquez Carmona, Dr. Jarrett Moyer
Spring 2023	Molecules, Genes & Cells (Amherst College) Instructors: Dr. Alexandra Purdy, Dr. Katerina Ragkousi, Dr. Christina Bowers

Grader

Spring 2026	Graph Theory (Amherst College) Instructor: Dr. Miriam Kuzbary
Fall 2025	Linear Algebra (Amherst College) Instructor: Dr. Joseph Palmer
Fall 2024	Introduction to Analysis (Amherst College) Instructor: Dr. Ryan Alvarado

SERVICE

Manuscript Peer Review

Journal of Computational Biophysics and Chemistry

Service to the Department and University

2026	Vice President, International Student Association, Amherst College
2024–2026	Founder & President, Biotechnology Club, Amherst College
2025	Organizer, Syllabus Design Workshops for Faculty, Amherst College
2023–2025	Pedagogical Fellow, Center for Teaching and Learning, Amherst College
2023–2025	Academic Coach, Strategic Learning Center, Amherst College
2023	Diversity, Equity, and Inclusion Fellow, Biology Steering Committee, Amherst College
2023	Panelist, STEM Research Invitational, Amherst College
2023	Student Panelist, Provost’s Annual Retreat on Teaching and Learning, Amherst College

Service to the Broader Community

2018–present	Member of Board, Talent Nurturing Program (“Chuong trinh Nuoi Duong Nhan Tai”)
2019–2022	Founder & President, Biology For All Vietnam Project
2021–2022	Project Lead & English Teacher, Passerelles Numériques Vietnam
2021	Organizer, International Conference on Raising Cybersecurity Awareness among Teenagers, Royal Thai Consulate-General
2020–2021	Admissions Committee & Mentor in Biology, Vietnam Math and Science Summer Program